Nguyen Thanh Duy Tan

AI Engineer Intern

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Career Objective

Short-term: Apply AI/ML knowledge to real projects and improve practical engineering skills.
Long-term: Become an AI Engineer capable of building reliable ML and LLM-based systems.

Education

Ho Chi Minh City University of Natural Resources and Environment Oct 2023 – Present
Major: Information Technology GPA: 3.0/4.0

Achievements

Academic Encouragement Scholarship – Good Classification, Academic Year 2024–2025.

Skills

- **Machine Learning:** Scikit-learn, Pandas, NumPy, Imbalanced Data Handling, EDA.
- **Generative AI & NLP:** LLMs, LangGraph, LangChain, Multi-Agent Systems, RAG.
- **MLOps & Software Engineering:** FastAPI, Docker, CI/CD (GitHub Actions), RESTful APIs.
- **Database & Storage:** MongoDB (Motor Async).
- **Tools:** Git, GitHub, Jupyter Notebook.
- **Soft Skills:** Problem-solving, system design thinking, self-researching technical docs.

Projects

VietLex - Vietnamese Legal RAG & Evaluation System **07/2026 - Present**

- Architected a legal RAG system over a pinned corpus of 518,255 Vietnamese legal documents, with Vertex/Qdrant v3 as the default structural retrieval path and explicit fail-closed routing.
- Built 1,024-dimensional Gemini dense-sparse retrieval with IDF and raw RRF over 51,801 Qdrant points spanning 4,969 audited documents; packaged matching SQLite/Zstandard full text and FTS5 for full-document pages, title/number search, and sparse calibration.
- Achieved 53/53 Document Recall@3, 30/30 Article Recall@3, 13/14 Clause Recall@3, and 39/40 all-required coverage on Golden-50, with zero retrieval errors or no-candidate cases.
- Completed 50/50 grounded generations, NeMo checks, and opt-in Ragas audits (0.864 Faithfulness, 0.920 Answer Accuracy, 0.870 Context Precision, 0.937 Context Recall) with zero judge errors at 5.77s P50 latency; validated by 915 passing tests.

Tech Stack: Python, FastAPI, Google Vertex AI, Qdrant Cloud, SQLite/FTS5, Zstandard, MongoDB, NeMo Guardrails, Ragas, Pytest, Docker, GitHub Actions.

GitHub: github.com/TanNguyen234/VietLex-Tech-Spec

CV AI Evaluation System – Multi-Agent LLM Application 03/2026 – 05/2026

- Built a LangGraph workflow for PDF extraction, candidate profiling, dynamic rubric generation, Tavily-based market enrichment, and parallel execution of six specialized evaluator nodes.
- Implemented deterministic cross-agent validation, meta-evaluation, optional job-description matching, and retry policies for transient LLM failures across configurable Gemini and Qwen providers.
- Developed an async FastAPI service with SSE progress streaming, MongoDB/Motor persistence, LLM-based CV validation, per-IP daily limits, unit/integration tests, and Docker deployment on Hugging Face Spaces.

Tech Stack: Python, FastAPI, LangGraph, LangChain, Gemini, Qwen, Tavily, MongoDB/Motor, PyMuPDF4LLM, Docker.
GitHub: github.com/TanNguyen234/CV-Review-System | **Demo:** foxy0505-ai-cv-reviewer.hf.space/app

Fraud Detection API & CI/CD Pipeline 05/2026 – 06/2026

- Trained an XGBoost classifier on the ULB credit-card dataset (approximately 0.17% fraud) using duplicate removal, time transformation, RobustScaler, stratified splitting, and class weighting with `scale_pos_weight`.
- Evaluated the model with PR-AUC and class-wise metrics, then served the preprocessing and model artifacts through versioned FastAPI/Pydantic endpoints for single and batch inference of up to 100 transactions.
- Added unit/integration tests and a GitHub Actions workflow that runs Ruff and Pytest before building and pushing the Docker image to Docker Hub.

Tech Stack: Python, XGBoost, Pandas, Scikit-learn, FastAPI, Pydantic, Pytest, Docker, GitHub Actions.
GitHub: github.com/TanNguyen234/End-to-End-Fraud-Detection-Pipeline